

1. Administrative & Study Identification

Full Study Title: A Phase I Randomised, Single-blind, Placebo-controlled Study to Assess the Safety, Tolerability, and Pharmacokinetics of AZD1613 Following Multiple Ascending Dose Administration in Participants With Autosomal Dominant Polycystic Kidney Disease **Short Title/Acronym:** PIONEER-PKD **Protocol Number:** D9050C00002 **NCT Number:** NCT07228364 **Sponsor Name:** AstraZeneca **Investigational Product:** AZD1613 (Monoclonal antibody / PAPPA-1 modulator) **Phase of Development:** Phase I **Medical Monitor:** AstraZeneca Clinical Operations Lead / Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the safety, tolerability, and pharmacokinetics of AZD1613 following multiple ascending dose administration in participants with autosomal dominant polycystic kidney disease (ADPKD). **Secondary Objectives:** To characterise the pharmacokinetics (PK) profile of AZD1613 and evaluate pharmacodynamic (PD) biomarkers of PAPPA-1 inhibition following subcutaneous or intravenous administration.

2.2 Background & Rationale To address the need for novel therapies targeting the underlying mechanisms of cyst growth in Autosomal Dominant Polycystic Kidney Disease (ADPKD). Unlike standard treatments that focus on managing symptoms, AZD1613 targets and interferes with the Pregnancy-associated plasma protein-A (PAPPA-1) pathway responsible for cyst growth, potentially offering a new approach to improving kidney function. Inclusion of a placebo arm is appropriate for benchmarking the safety and tolerability of AZD1613 in this patient population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Single-blind **Randomization:** Randomised (Parallel Assignment)

3.2 Planned Enrollment & Duration Target \$N\$: 40 participants **Number of Sites:** Global (USA, China, United Kingdom) **Estimated Study Start:** 10-Nov-2025 **Estimated Primary Completion:** 04-Jan-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Patients with ADPKD Mayo Class (IB-IE), as per clinical diagnosis (MIC) assessed centrally. 2. Confirmed eGFR = 45 to 90 mL/min/1.73m². 3. Body weight $\geq$ 45 kg and body mass index within the range 18 to 35 kg/m² (inclusive). Females are to be of non-childbearing potential.

4.2 Exclusion Criteria 1. Any evidence of cardiac, vascular, and other renal conditions which, in the investigator's opinion, makes it undesirable for the participant to participate in the study. 2. Positive hepatitis C antibody, hepatitis B virus surface antigen, or human immunodeficiency virus (HIV) test at screening. 3. Kidney cyst interventions such as cyst aspiration or cyst fenestration within 12 weeks prior to screening and during the screening period, or anticipated during follow-up.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: 4 doses of AZD1613 or placebo administered on Days 1, 29, 57, and 85 (Multiple Ascending Dose methodology). **Route of Administration:** Subcutaneous (SC) or Intravenous (IV). **Duration of Treatment:** ~3 months (85 days) of active dosing, with off-treatment safety follow-up evaluated through Day 189.

5.2 Concomitant & Prohibited Therapy Permitted: Stable treatment regimen of antihypertensive therapy for at least 30 days prior to the screening visit. **Prohibited:** Medications associated with QT prolongation that require discontinuation; recent or planned kidney cyst interventions.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -1	Informed Consent, Medical History, ECG, Labs, Viral screening
Baseline	Day 1	Randomization, First Dose (AZD1613 or Placebo), PK/PD sampling
Interim	Days 29, 57, 85	Subsequent Doses, Safety Labs, PK/PD Assessments
End of Study	Day 189 $\pm$ 3	Final Safety Evaluation, PK/PD sampling, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence and severity of Adverse Events (AEs) and Serious Adverse Events (SAEs) to assess overall safety and tolerability. **Measurement Method:** Clinical evaluation, vital signs, ECGs, and safety laboratory tests throughout the study up to Day 189.

7.2 Secondary & Exploratory Endpoints 1. Pharmacokinetics (PK) parameters of AZD1613 (e.g., maximum observed concentration [C_{max}] and area under the curve [AUC]). 2. Changes in pharmacodynamic (PD) biomarkers of PAPA-1 inhibition.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via clinical assessments, investigator interviews, and laboratory/ECG monitoring. **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal safety review committee managed by AstraZeneca.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Analysis Set for safety data; PK Analysis Set for pharmacokinetic parameters. **Sample Size Justification:** Target N = 40 is considered appropriate for a Phase I multiple ascending dose study to evaluate safety, tolerability, and PK profiles without formal statistical hypothesis testing. **Handling of Missing Data:** Missing PK/PD data generally not imputed in Phase I unless otherwise specified; Last Observation Carried Forward (LOCF) for longitudinal safety data if applicable.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to ensure protocol adherence, patient safety, and centralized diagnosis validation. **Audit Trail:** Details on eCRF locking and data clarification forms managed via standard clinical data systems.

11. Study Identifiers & Governance

Primary ID: D9050C00002 **Registry Number:** NCT07228364 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** PIONEER-PKD **Official Regulatory Title:** A Phase I Randomised, Single-blind, Placebo-controlled Study to Assess the Safety, Tolerability, and Pharmacokinetics of AZD1613 Following Multiple Ascending Dose Administration in Participants With Autosomal Dominant Polycystic Kidney Disease

12. Clinical Framework (ADPKD Specific)

Primary Condition: Autosomal Dominant Polycystic Kidney Disease (ADPKD) **Diagnosis Threshold:** ADPKD Mayo Class (IB-IE) and eGFR 45 to 90 mL/min/1.73m2 **Medical Methodology:** Pregnancy-associated plasma protein-A (PAPPA-1) targeted modulation **Optimization Target:** Safety, tolerability, and optimal PK/PD profile evaluation

13. Study Procedures & Methodology

Management Pathway: Multiple Ascending Dose (MAD) Phase 1 clinical pathway **Education Protocol (HCP):** Investigator protocol training, centralized ADPKD Mayo Class assessment procedures, and strict safety reporting guidelines **Education Protocol (Patient):** Informed consent, schedule orientation, and counseling for non-childbearing requirements **Quality Audit Mechanism:** Centralized assessment of clinical diagnosis (MIC) and blinded data review

14. Observation & Follow-Up Schedule

Total Duration: Follow-up through Day 189 (approx. 27 weeks)
Onsite Visit Cadence: Days 1, 29, 57, and 85 for active dosing, followed by specific safety evaluation visits
Remote Monitoring Frequency: As per standard Phase 1 study remote monitoring guidelines
Checklist Review Interval: Continuous tracking of AEs and concomitant medications during the entire 189-day observation period

15. Sign-off & Approval

Role	Name	Signature	Date	---	---	---	---
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]				
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]				